SIADS 696

Milestone II

Fall 2026

Project

Who Gets Left Out?
Predicting and Profiling
Financial Exclusion in New York City

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Project Overview

We examine whether socioeconomic conditions predict physical banking access and whether New York City neighborhoods form distinct financial health profiles. The unit of analysis is the 55 NYC Public Use Microdata Areas (PUMAs). Physical access is measured as FDIC-insured bank and credit union branches per 10,000 residents; this neighborhood-level proxy does not capture travel time, service quality, hours, affordability, or digital access.

For supervised learning, median income, poverty, labor-force participation, high school graduation, and homeownership predict branch density. Ridge regression, random forest, and K-nearest neighbors (KNN) are compared with shuffled five-fold cross-validation using R^2, MAE, and RMSE. KNN performs best overall, but modest explanatory power and large errors for unusual PUMAs show that socioeconomic variables alone do not fully explain branch locations.

For unsupervised learning, K-means and Ward hierarchical clustering are evaluated for two to six clusters using 13 standardized socioeconomic, demographic, financial health, and branch access indicators. Ward with two clusters produces the strongest separation: a larger, lower-income and lower access profile and a smaller, higher-income and higher access profile. Held-out bank and credit union account ownership is higher in the smaller cluster, although the difference is descriptive, not causal. PCA-based assignments are stable, while removing race composition features changes some assignments.

The project extends prior NYC financial exclusion and banking desert research by combining neighborhood financial health indicators with geocoded branch locations and identifying multidimensional profiles rather than a binary desert designation. Results can guide further equity-focused investigation, outreach, and planning. However, they are exploratory, aggregate, and noncausal and should not determine individual or high stakes decisions.

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GitHub Repository: https://github.com/sbabtiwa/who-gets-left-out.git

Related Work

Ratcliffe, Caroline, et al. "Where Are the Unbanked and Underbanked in New York City?" Urban Institute, 2015.
https://www.nyc.gov/assets/dca/CitywideFinancialServicesStudy/Research-Brief.pdf

This study examines the geographic distribution of unbanked and underbanked households across New York City and analyzes how financial exclusion varies with socioeconomic and demographic characteristics. It is especially relevant because it supports the Neighborhood Financial Health Digital Mapping and Data Tool used in our project.

Our proposed analysis extends this work by adding FDIC bank and credit union branch locations and applying PCA, hierarchical clustering, K-means, and spatially constrained clustering to identify multidimensional neighborhood profiles.

Federal Reserve Bank of St. Louis. "Banking Deserts Become a Concern as Branches Dry Up." *The Regional Economist*, 2017.
https://www.stlouisfed.org/publications/regional-economist/second-quarter-2017/banking-deserts-become-a-concern-as-branches-dry-up

This article examines the emergence of banking deserts as physical bank branches close, particularly in lower-income and rural communities. It explains how geographic access to bank branches can remain important for households and businesses despite the growth of online banking.

Our project differs by focusing specifically on New York City and combining physical branch access with PUMA-level measures of unbanked status, underbanked status, prepaid card use, poverty, unemployment, and income. Rather than applying only a banking desert designation, we will identify multiple types of neighborhood financial environments through clustering.

Morgan, Donald P., Maxim L. Pinkovskiy, and Bryan Yang. "Banking Deserts, Branch Closings, and Soft Information." Federal Reserve Bank of New York, *Liberty Street Economics*, March 7, 2016.
https://libertystreeteconomics.newyorkfed.org/2016/03/banking-deserts-branch-closings-and-soft-information/

This study examines the location of banking deserts and considers whether bank branch closures disproportionately affect lower-income and economically vulnerable communities. It also discusses why physical branches remain important when financial relationships rely on local or "soft" information that may not be captured through automated systems.

Our project builds on this study by shifting the emphasis from the consequences of branch closures to the identification of neighborhood classification. We will examine whether PUMAs with comparable socioeconomic conditions exhibit different levels of banking access and identify unusual cases, such as neighborhoods with many branches but persistent underbanking. Comparing conventional and spatially constrained clustering will also show whether these financial profiles follow geographic boundaries or occur across nonadjacent parts of the city.

Data Sources

This analysis combines neighborhood financial health indicators, FDIC-insured bank and credit union locations, Census geocoding and PUMA boundary resources, and Census population estimates. The local source files are stored in the project directory. The primary analysis geography is the 55 New York City Public Use Microdata Areas (PUMAs); bank and credit union locations are geocoded and assigned to those PUMAs so that branch access can be compared with neighborhood characteristics.

Neighborhood Financial Health Digital Mapping and Data Tool: The source is the NYC dataset listed in the Data.gov catalog under "Neighborhood Financial Health Digital Mapping and Data Tool;" the project uses the downloaded local CSV file neighborhood_financial_health.csv. The source file has 385 rows and 52 columns, with records for 55 PUMAs and fields including Year Published, PUMA, Borough, Neighborhoods, NYC_Poverty_Rate, Median_Income, demographic percentages, Goal, TotalOutcome, IndexScore/ScoreRank, and indicator outcomes and ranks. The time period is 2012 to 2020. Rows are organized by PUMA and goal and also include summary categories; after excluding "Overall Index" and "Majority Race/Ethnicity," the analysis retains 275 rows (55 PUMAs × five financial health goals).

FDIC Bank Locations: The source is the FDIC Bank Locations dataset listed in the Data.gov catalog; its downloaded local file is locations.csv, a CSV table with 78,075 nationwide location records and 30 columns. Important fields include UNINUM (branch/location identifier), CERT (institution identifier), ADDRESS, CITY, COUNTY, STALP, ZIP, NAME, and OFFNAME. Filtering to New York State yields 4,005 records; mapping the five NYC counties to boroughs yields 1,404 NYC branch records. The source includes a RUNDATE field (the displayed records show 09/04/2026); it is a branch location snapshot, not a multi-year time series. Branches are counted by distinct UNINUM, and institutions by distinct CERT.

To handle census geocoding and geographic boundaries, the 1,404 NYC branch records from the FDIC Bank Locations dataset were submitted to the U.S. Census Bureau Batch Geocoder using census_batch_input.csv (CSV, no header; UNINUM, street, city, state, ZIP), and the downloaded GeocodeResults.csv was parsed as CSV. Results contain match status, match type, matched address, longitude/latitude, and Census tract/block identifiers: 1,365 were Match, 12 Tie, and 27 No_Match. The spatial reference is the Census TIGER/Line 2019 New York PUMA shapefile, downloaded as tl_2019_36_puma10.zip (a zipped shapefile); it contains 145 New York State polygons and uses the 2010 PUMA geography. Fields used include PUMACE10/GEOID10 and polygon geometry. Matching to the financial-health PUMA codes retains 55 NYC polygons, and all 1,365 successfully geocoded NYC points were assigned to a PUMA.

Credit Unions: The data come from the National Credit Union Administration (NCUA) and the analysis of Credit Union Corporate Call Report Data page, using its "List of Active Federally Insured Credit Unions" download. The project file, `FederallyInsuredCreditUnions_2026q2.xlsx`, is an Excel workbook read from the 'FICUs' sheet into a panda DataFrame. It contains 4,214 active federally insured credit unions nationally and 23 columns. The 'Year and quarter' field identifies the snapshot as 2026 Q2 (shown as 2026.2); this is a single-quarter listing, not a multi-period time series. Important fields include 'Charter number' (the credit union identifier used for counting), 'Credit Union name', 'Credit Union type', 'NCUA region', and 'Low-income designation'; mailing address fields for street, city, state, and ZIP; and institution measures such as 'Members', 'Total assets', 'Total deposits', return on average assets, net worth ratio, loan-to-share ratio, and four-quarter growth rates for deposits, loans, assets, members, and net worth. The addresses are mailing addresses, so they should not automatically be interpreted as physical branch locations.

Population API: The Census Bureau's 2020 ACS 5-year API was queried for New York State PUMA geographies. The request returned JSON for NAME and B01003_001E (estimated total population); the response was converted to a pandas table, and the matched NYC subset was saved as acs_puma_population_2020.csv. The analysis uses 55 records, one per study PUMA, with PUMA, ACS_PUMA_Name_2020, and ACS_Population_2020. B01003_001E is from Detailed Table B01003; the 2020 ACS 5-year estimates represent 2016–2020 and are used as denominators for branches per 10,000 residents.

The prepared analysis files include the goal-level joined CSV (neighborhood_financial_health_with_banks.csv; 275 records) and the flattened PUMA-level CSV (neighborhood_financial_health_flat_by_puma.csv; 55 records, one per PUMA). The modeling table uses the 55 PUMAs as observations, combining socioeconomic indicators with FDIC branch counts and ACS population, and derives FDIC branches per 10,000 residents. The sources do not form a longitudinal panel: the neighborhood indicators and population estimates are 2020 based, while the FDIC file is a dated branch snapshot.

Feature Engineering

Feature engineering integrated the Neighborhood Financial Health (NFH), FDIC Bank Locations, credit union, and American Community Survey (ACS) population data into one record for each of New York City's 55 Public Use Microdata Areas (PUMAs). Geographic identifiers were standardized as four-character strings before sources were joined.

The NFH data contained five financial health goals and two summary categories for each PUMA. We removed the Overall Index and Majority Race/Ethnicity rows, retained Build Assets, Financial Services, Financial Shocks, Goods and Services, and Jobs and Income, standardized the PUMA and Join fields, and verified that every PUMA had one row for each retained goal.

Each PUMA was classified according to the published NFH methodology. A PUMA was labeled **NFH Advantaged** if it ranked in the top third for at least four of the five goals, **NFH Vulnerable** if it ranked in the bottom third for at least four goals, and **Variable** otherwise. The NFH data were then reshaped from five goal-level rows per PUMA into one wide row per PUMA. Goal-specific fields received names such as Financial Services__IndexScore, while shared neighborhood characteristics were retained only once.

The national FDIC Bank Locations file was first filtered to New York State and then restricted to New York City's five counties: Bronx, Kings, New York, Queens, and Richmond. County names were mapped to borough names, ZIP codes were stored as five-character strings, and duplicate locations were removed using UNINUM, the FDIC branch location identifier. CERT, the FDIC certificate number, was retained to identify distinct banking institutions.

FDIC addresses were batch-geocoded by the U.S. Census Bureau and merged to the source records by UNINUM. Tied or unmatched records without reliable coordinates and pointed outside a broad New York City bounding box were excluded from spatial analysis. Valid coordinates were assigned to 2010 PUMA polygons, then aggregated into Bank_Branch_Count (distinct UNINUM locations) and Unique_Bank_Count (distinct CERT institutions).

Credit union records used a unique row-level location identifier because one charter number can represent multiple locations. Addresses were geocoded through the Census Batch Geocoder. Ties were retried with the Census single address service, unmatched addresses were cleaned of suite, floor, and unit information, and remaining cases were queried through OpenStreetMap Nominatim. ZIP and city centroids documented unresolved mailing addresses but were excluded from branch access measures. Precisely geocoded locations were assigned to PUMAs to create Credit_Union_Count, which was combined with bank locations as Bank_Credit_Union_Branch_Count.

Population estimates came from the 2020 ACS five-year Detailed Table B01003, variable B01003_001E. PUMA codes were standardized and matched to the 55 PUMAs in the prepared NFH dataset. The population data were checked to ensure that each PUMA appeared once, every NFH PUMA had a matching population record, and all estimates were present and positive. The resulting feature was named ACS_Population_2020.

To compare PUMAs of different sizes, we calculated FDIC bank, credit union, and combined branch rates per 10,000 residents by dividing each count by ACS_Population_2020 and multiplying by 10,000. These measures differ from the original NFH bank density indicator, which used 2018 bank and credit union records and ACS 2012–2016 population estimates.

We preserved leading zeros in PUMA and ZIP codes and removed duplicate FDIC locations by UNINUM. We used row-level identifiers for credit unions and excluded out-of-area or imprecisely geocoded records from branch counts. ZIP and city centroids were not treated as branch locations. Missing PUMA assignments were reviewed before absent branch counts were set to zero. Numeric fields were converted to numeric types. The remaining missing values were imputed within each training fold to prevent leakage.

For supervised learning, the target variable was FDIC_Branches_per_10k, representing the number of FDIC-insured bank branches per 10,000 residents. Five socioeconomic predictors were selected: median income, poverty rate, labor force participation, high school graduation, and homeownership. These features were used to predict differences in physical banking access across the 55 NYC PUMAs.

For unsupervised learning, only numeric variables were used. These included poverty rate, median income, demographic percentages, bank and credit union access measures, and, where appropriate, the five NFH goal index scores. NFH_Classification with the three classes NFH Advantaged, Variable, and NFH Vulnerable were created. Variables that directly determined the NFH_Classification value, such as goal-specific score tiers, top-third goal counts, bottom-third goal counts, and the final classification rule, were excluded to prevent target leakage. Continuous variables were standardized before applying scale-sensitive methods such as k-means clustering or principal component analysis. Geographic and descriptive variables, including PUMA, borough, neighborhood name, community district, and NFH classification, were excluded from the clustering inputs but retained for interpreting the resulting clusters. To avoid placing excessive weight on access to financial institutions, normalized branch measures were preferred to include every raw and normalized count simultaneously.

The final dataset contained geographic and descriptive fields such as PUMA, Borough, Neighborhoods, CD, and the ACS PUMA name; socioeconomic and demographic features such as NYC_Poverty_Rate, Median_Income, Perc_White, Perc_Black, Perc_Asian, Perc_Other, Perc_Hispanic, and ACS_Population_2020; and financial-institution features such as Bank_Branch_Count, Unique_Bank_Count, Credit_Union_Count, Bank_Credit_Union_Branch_Count, and their corresponding rates per 10,000 residents. It also included NFH_Classification and the goal-specific total outcomes, ranks, index scores, score tiers, indicator values, indicator outcomes, indicator ranks, and indicator definitions for Build Assets, Financial Services, Financial Shocks, Goods and Services, and Jobs and Income. The complete feature list, final_feature_list.csv, is available in the data directory in our GitHub repository and is also available in the last cell of the Python notebook.

Part A. Supervised Learning

Supervised Learning Workflow

The goal of the supervised learning analysis was to determine how well socioeconomic characteristics predict physical banking access across NYC PUMAs. Physical banking access was represented by the number of FDIC-insured bank branches per 10,000 residents.

The dataset contained 55 NYC PUMAs, with FDIC_Branches_per_10k as the target variable and five socioeconomic predictors: median income, poverty rate, labor force participation, high school graduation, and homeownership.

Before modeling, we examined the data through descriptive statistics, correlations, structure, and quality. Within the 55 observations, and only one PUMA had a missing labor force participation value, which we retained due to the small dataset and handled using median imputation within each cross-validation training fold.

For Ridge Regression and KNN, the socioeconomic variables were standardized within the modeling pipelines so that differences in numerical scale would not disproportionately influence the models. This was particularly important for KNN because it relies on distances between observations to make predictions.

Learning Methods

We compared three regression models with different approaches to predicting physical banking access:

Ridge Regression: A regularized linear model that provided an interpretable baseline. Ridge was useful because several socioeconomic predictors were correlated and regularization helps reduce overfitting when predictors contain overlapping information.

Random Forest Regression: A tree-based model that captures nonlinear relationships and interactions between socioeconomic predictors.

K-Nearest Neighbors (KNN) Regression: A distance-based model that predicts banking access using PUMAs with similar socioeconomic characteristics.

Using these three model families allowed us to compare models with different assumptions: a regularized linear model, a nonlinear tree-based model, and a distance-based model.

Hyperparameter Exploration and Cross-Validation

Because the dataset contains only 55 PUMAs, relying on a single train/test split could produce an unstable estimate of model performance. Therefore, we used 5-fold cross-validation with the same data splits across all three model families to ensure a consistent comparison.

Model performance was primarily evaluated using Root Mean Squared Error (RMSE). We turned each model's hyperparameters using cross-validation. The best configurations were Ridge with alpha = 10, Random Forest with 200 trees, a maximum depth of 5 and minimum of 4 samples per lead, and KNN with 3 distance-weighted neighbors. For KNN, we tested k values of 3, 5, 7, 9, and 11.

Supervised Evaluation

Model Performance

Because banking access is a continuous variable, we evaluated models using three regression metrics:

RMSE: Measures prediction error while penalizing larger errors more heavily.

MAE: Measures the average absolute prediction error.

R^2: Measures how much variation in banking access the model explains.

Table A1 below compares the best-performing model from each model family using the mean and standard deviation of the evaluation metrics across the 5 cross-validation folds.

Table A1. Cross-validated model performance

Model	R2 Mean +/- SD	MAE Mean +/- SD	RMSE Mean +/- SD
Ridge Regression	0.109 +/- 0.529	0.74 +/- 0.234	1.15 +/- 0.695
Random Forest	0.164 +/- 0.5	0.744 +/- 0.21	1.153 +/- 0.715
KNN	0.39 +/- 0.283	0.693 +/- 0.265	0.999 +/- 0.624

KNN had the strongest overall performance across the three model families. It produced the highest mean R^2, the lowest mean MAE, and the lowest mean RMSE. Therefore, it was selected for further evaluation.

Feature Importance

KNN does not provide built-in feature importance scores. Therefore, permutation importance is used to evaluate how much each predictor contributes to the model's predictions. Permutation importance randomly shuffles one feature at a time and measures how much model performance decreases. A larger decrease in performance indicates that the feature is more important to the model.

Table A2. Permutation importance results

Removed Feature	Mean RMSE
Median_Income	0.334
Poverty_Rate	0.2784
Labor_Force_Participation	0.1582
High_School_Graduation	0.076
Homeownership	0.066

Median income had the highest permutation importance, followed by high school graduation and labor force participation. This suggests that KNN relies most heavily on these characteristics when predicting physical banking access. Poverty rate and homeownership had lower permutation importance, suggesting that they contributed less unique predictive information to the KNN model. This may be due to correlations among the features.

Ablation Analysis

An ablation analysis was performed by removing one predictor at a time and reevaluating KNN using 5-fold cross-validation. If removing a feature increases RMSE, the feature is contributing useful predictive information. If removing a feature decreases RMSE, model performance improves without that feature.

Table A3. KNN ablation analysis

Removed Feature	Mean RMSE
None (All Features)	0.9986068532
Median_Income	1.009431966

Poverty_Rate	1.042234791
Labor_Force_Participation	1.051286673
High_School_Graduation	1.177374401
Homeownership	0.9361106919

Removing High School Graduation caused the largest increase in RMSE, from approximately 0.999 to 1.177. This suggests that high school graduation provides important predictive information for the KNN model. Removing Labor Force Participation and Poverty Rate also increased RMSE, while removing Median Income caused only a small increase. In contrast, removing Homeownership decreased RMSE to approximately 0.936, indicating that KNN performed slightly better without this feature. Homeownership may therefore provide limited additional predictive information or introduce noise into the model. The ablation results do not perfectly match the permutation importance ranking. This may reflect correlations among the socioeconomic predictors.

Sensitivity Analysis

A sensitivity analysis was performed to examine how KNN performance changes with the number of neighbors (k). Smaller values of k make predictions based on fewer, more similar PUMAs, while larger values use a broader group of neighbors. The model was evaluated across several values of k using the same 5-fold cross-validation procedure. RMSE was used to compare performance, with lower RMSE indicating better predictive performance.

Table A4. KNN sensitivity analysis by number of neighbors

Number of K Neighbors	Mean RMSE
3	0.999
5	1.105
7	1.085
9	1.104
11	1.099

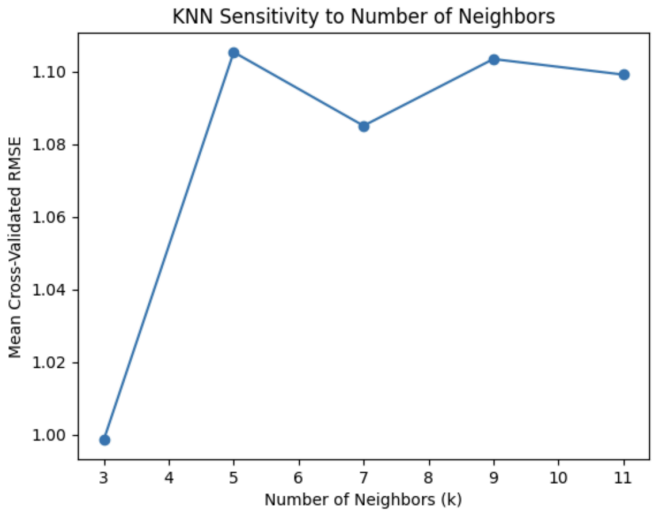


Figure A1. KNN Sensitivity to Number of Neighbors.

The sensitivity analysis shows that KNN performed best when using three neighbors, with a mean cross-validated RMSE of approximately 1.00. Increasing the number of neighbors from 3 to 5 increased prediction error. For values of k between 5 and 11, RMSE remained relatively stable at approximately 1.09 to 1.11.

These results suggest that the model performs better when predictions are based on a small number of highly similar PUMAs. However, performance is relatively stable across the larger values of k that were tested.

Model Tradeoffs

The results highlight several tradeoffs among the supervised learning approaches. KNN achieved the strongest predictive performance, but it is less directly interpretable than Ridge Regression because it does not produce coefficients that describe the direction and magnitude of each predictor's relationship with banking access.

KNN also depends on the choice of the number of neighbors. Using a small value of k allows the model to capture similarities among closely related PUMAs, but it may make predictions more sensitive to individual observations. Larger values of k produce smoother predictions but may overlook meaningful local differences.

There is also a tradeoff between including more socioeconomic features and model simplicity. The ablation analysis showed that removing Homeownership slightly improved RMSE, suggesting that adding more predictors does not necessarily improve predictive performance.

Failure Analysis

To better understand where KNN struggled, we examined the PUMAs with the largest absolute difference between predicted and observed baking access. The three PUMAs with the largest prediction errors were 3807, 3810, and 4103.

Table A5. Largest KNN prediction errors

PUMA	Actual Branches	Predicted Branches	Absolute Error
3807	11.23	4.26	6.97
3806	1.78	3.67	1.89
4013	2.46	1.06	1.40

PUMA 3807 was the largest failure. Its actual branch density was 11.23 branches per 10,000 residents, while the KNN model predicted only 3.01. Although PUMA 3807 has relatively high income and labor force participation and relatively low poverty, these characteristics did not distinguish it as having an unusually high branch density. This suggests that important factors related to bank branch availability may not be captured by the predictors used in the model.

PUMA 3810 also had its branch density underestimated, with an actual value of 5.26 compared with a prediction of 3.84. This PUMA had the highest median income in the dataset and relatively low poverty, suggesting that the model may not have fully captured the relationship between higher-income neighborhoods and branch availability.

PUMA 4103 had an actual branch density of 2.88 compared with a prediction of 1.50. It had lower median income, higher poverty, and lower labor force participation than the other two failure cases. The model again underestimated the observed branch density.

Overall, the failure analysis suggests that KNN can struggle when a PUMA has a branch density that is substantially different from what would be expected based on its socioeconomic characteristics. This may occur because the current predictors do not capture all factors that influence the geographic availability of bank branches.

Part B. Unsupervised Learning

For Unsupervised Learning, we used the same dataset as in Supervised Learning. The analysis grouped 55 NYC PUMAs using 13 standardized features spanning poverty rate, median income, race/ethnicity composition, selected Neighborhood Financial Health indicators, and FDIC bank/credit union branch density per 10,000 population. One missing value from 13 standardized features was median-imputed. We intentionally held out Financial Services__Ind5Outcome (Bank/Credit Union Utilization) so it could serve as an external validation outcome later.

The analysis compares K-means, which partitions observations around cluster centers, with Ward hierarchical clustering, which creates nested groups by merging observations to limit within-cluster variation. For both methods, we tested cluster counts, k values, from 2 to 6 and assessed separation using silhouette, Calinski-Harabasz, and Davies-Bouldin scores. Both methods' highest silhouette scores occurred at two clusters. See Figure B1, Cluster Scores Table, and Figure B2, Best k Table, below for detail.

```
Unsupervised sample: 55 PUMAs; 13 standardized input features.
Financial Services__Ind5Outcome (Bank / Credit Union Utilization) is held out for external validation.
Input missing values imputed by feature median: 1
```

	Method	k	Silhouette	Calinski-Harabasz	Davies-Bouldin
0	K-means	2	0.282	25.882	1.350
2	K-means	3	0.261	24.921	1.283
4	K-means	4	0.258	23.696	1.243
6	K-means	5	0.244	21.255	1.323
8	K-means	6	0.229	19.457	1.314
1	Ward method	2	0.322	21.094	1.126
3	Ward method	3	0.239	22.550	1.275
5	Ward method	4	0.237	22.197	1.314
7	Ward method	5	0.231	20.026	1.272
9	Ward method	6	0.224	18.491	1.256

Figure B1. Cluster score table.

Best k within each method by silhouette; compare all indices and cluster profiles before interpretation.

	Method object	k int64	Silhouette float64	Calinski-Harabasz	Davies-Bouldin float64
0	K-means	2	0.282	25.882	1.35
1	Ward method	2	0.322	21.094	1.126

Figure B2. Best k table

Ward's method gave the stronger silhouette score of 0.322 compared with 0.282 for K-means. Its Davies–Bouldin score was also lower (1.126 vs. 1.35). Under the selected Ward solution, the 55 PUMAs split into a large group of 45 (cluster 0) and a smaller group of 10 (cluster 1). The modest silhouette values suggest the groups are overlapping neighborhood profiles, not sharply distinct types. To visualize results, we constructed K-means elbow curve, Ward hierarchical dendrogram, PCA projections, and Standardized feature profile heatmap for Ward method (k=2). See Figure B3 and B4.

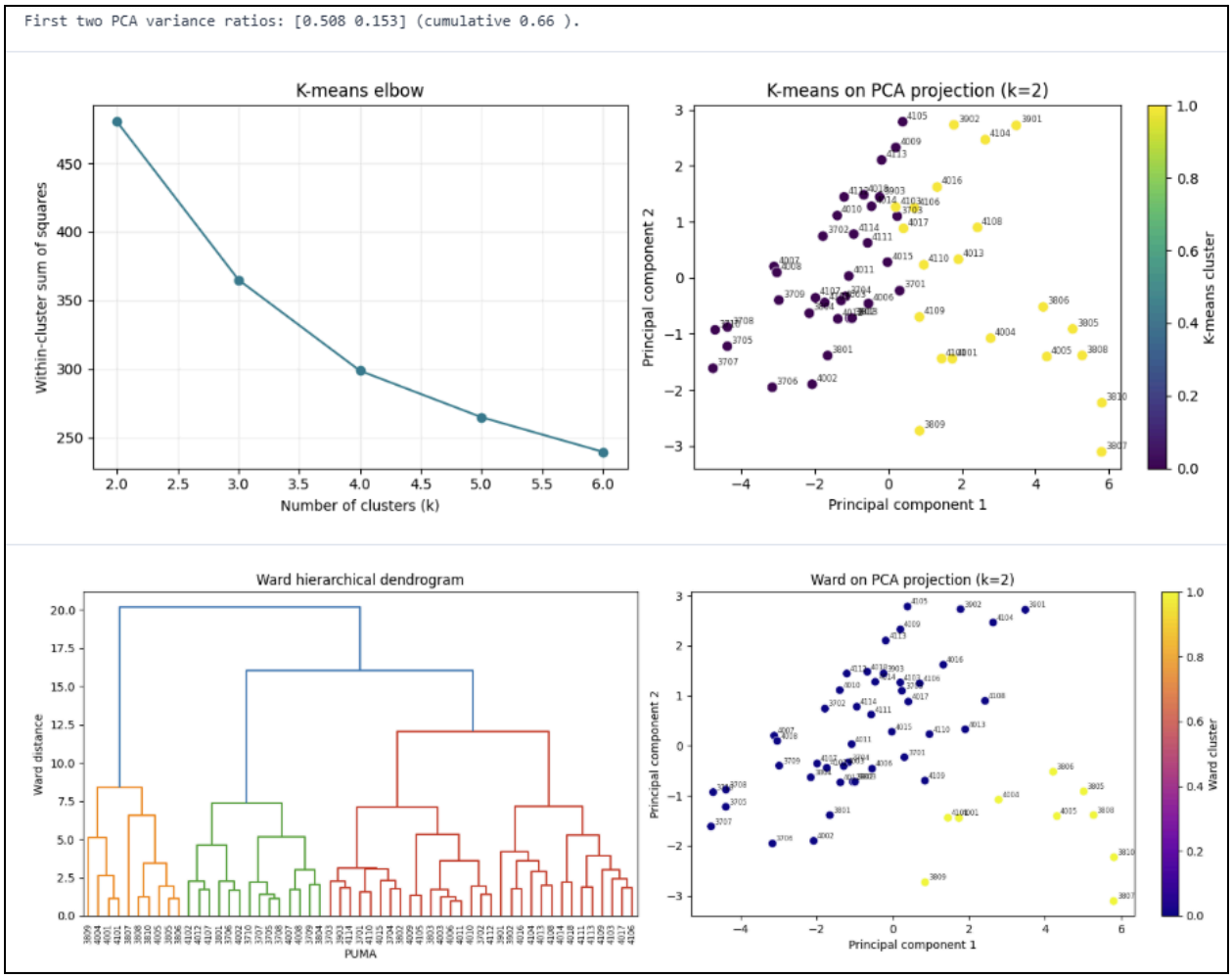


Figure B3. K-means and Ward clustering visualizations

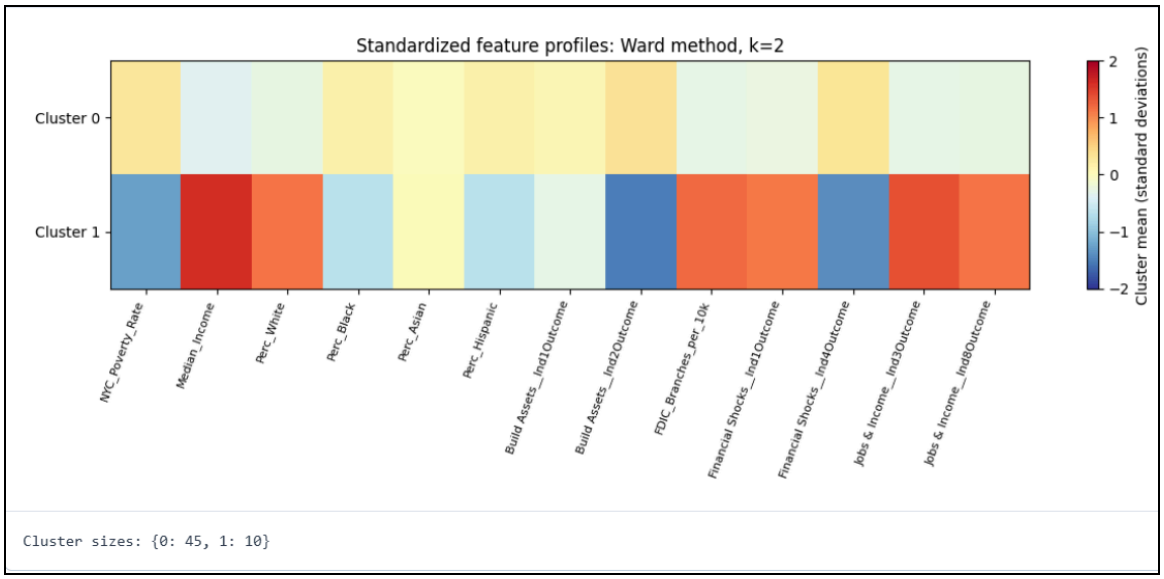


Figure B4. Ward standardized feature-profile heatmap.

Table B1, shown below, displays clear differences in socioeconomic conditions and access to banking between the two Ward clusters. Cluster 0 has a higher median poverty rate (22.0% compared with 10.4%) and a much lower median income ($25,920 compared with $61,618.50). It also has approximately half the median FDIC branch density of Cluster 1 (1.207 versus 2.335 branches per 10,000 residents). Cluster 1 also performs better on the financial-shock indicator (0.73 versus 0.52) and the jobs-and-income indicator (0.936 versus 0.804). This indicates stronger financial resilience and better employment and income outcomes for cluster 1. However, the build-assets indicator shows similar results across the two clusters, with Cluster 0 having a slightly higher median (0.316 versus 0.302).

Overall, Cluster 0 represents neighborhoods with greater economic disadvantages and more limited physical access to banking. Cluster 1 represents more economically advantaged neighborhoods with greater branch access and stronger financial-health outcomes. These findings describe patterns across PUMAs and should not be interpreted as rankings or as descriptions of individual residents.

Input Features	Cluster 0	Cluster 1
NYC_Poverty_Rate	0.22	0.104
Median_Income	25920.0	61618.5
FDIC_Branches_per_10k	1.207	2.335
Financial Shocks__Ind1Outcome	0.52	0.73
Build Assets__Ind1Outcome	0.316	0.302
Jobs & Income__Ind8Outcome	0.804	0.936

Table B1. Cluster medians for selected input features in original units.

External validation held out bank/credit-union utilization ('Financial Services__Ind5Outcome') from clustering. The smaller, higher-income/higher-access cluster had mean ownership 0.934 (median 0.940), versus 0.862 (median

0.878) in the larger cluster, a modest descriptive difference. This is consistent with the group's income and branch-density profiles. It does not establish that greater branch access causes higher account ownership. See Figure B5 for detail.

External validation: held-out bank/credit-union account ownership by cluster:

Cluster	count	median	mean
0	45	0.878	0.862
1	10	0.940	0.934

Figure B5. External validation using bank and credit union utilization.

Figure B6 highlights the distribution of boroughs for the two clusters. The smaller group, cluster 1, was concentrated in Manhattan, Brooklyn, and Queens: six PUMAs were in Manhattan, three in Brooklyn, and one in Queens. It included neither Bronx nor Staten Island PUMAs. Figure B7 lists all neighborhoods in cluster 1 along with their Financial Service indicator.

Borough counts by cluster:

Cluster	Borough	Bronx	Brooklyn	Manhattan	Queens	Staten Island
0		10	15	4	13	3
1		0	3	6	1	0

Figure B6. Borough counts by cluster.

PUMAs in the smallest cluster:

Visualize

	PUMA string	Borough str	Neighborhoods str	Cluster int64	Financial Services_Ind5Outcome float64
	3805 10% 3806 10% 8 others 80%	Manhattan 60% Brooklyn 30% Queens 10%	Upper East Side 10% Upper West Side & West Side 10% 8 others 80%	1 - 1	0.899 - 0.969
14	3805	Manhattan	Upper East Side	1	0.963
15	3806	Manhattan	Upper West Side & West Side	1	0.948
16	3807	Manhattan	Chelsea, Clinton & Midtown Business District	1	0.949
17	3808	Manhattan	Murray Hill, Gramercy & Stuyvesant Town	1	0.969
18	3809	Manhattan	Chinatown & Lower East Side	1	0.899
19	3810	Manhattan	Battery Park City, Greenwich Village & Soho	1	0.967
23	4001	Brooklyn	Greenpoint & Williamsburg	1	0.901
26	4004	Brooklyn	Brooklyn Heights & Fort Greene	1	0.909
27	4005	Brooklyn	Park Slope, Carroll Gardens & Red Hook	1	0.931
41	4101	Queens	Astoria & Long Island City	1	0.9

Figure B7. Neighborhoods in the smallest cluster

We performed sensitivity analysis to PCA representation by setting PCA's n_components parameter to 0.85 and exclusion of race-composition features. A sensitivity table, Exhibit B9, was constructed to compare clustering results across different data representations. We compared Adjusted Rand Index (ARI), clustering results to the original clustering on standardized raw features (ground truth). Sensitivity checks indicated Ward assignments were unchanged after PCA retaining 85.6% variance (ARI 1.000). Silhouette improved from 0.322 to 0.362. Removing race composition features changed the partition (ARI 0.732), so cluster membership is representation-sensitive. The elbow curve and tested k values show parameter sensitivity. No single internal score establishes a uniquely correct cluster count. The K-means and Ward visualizations, PCA projections, and standardized profile heatmap should be interpreted together with the stability and external validation results.

Sensitivity for Ward method, k=2; PCA retained 85.6% of variance in 4 components.

	Representation	Dimensions	Silhouette	ARI vs raw
0	Standardized raw features	13	0.322	1.000
1	PCA retaining 85% variance	4	0.362	1.000
2	Standardized features, race composition removed	9	0.463	0.732

Figure B8. Ward sensitivity analysis

In plain English, race-composition features influenced how the algorithm grouped neighborhoods. When those features were removed, some PUMAs were assigned to different clusters. The ARI of 0.732 means the new grouping still resembles the original in many ways, but it is not identical. It does not mean that exactly 73.2% of neighborhoods stayed in the same group.

The silhouette score increased from 0.322 to 0.463, meaning the groups were more clearly separated using the remaining features. That does not necessarily make the new grouping more accurate or preferable. It means the clusters are more internally distinct under that particular set of inputs. This analysis shows that neighborhood profiles are partly sensitive to including race composition. It does not establish that race composition causes differences in banking access or account ownership.

Novel contribution (Unsupervised Learning)

Compared with the NFHI methodology which uses predefined indicator scores, rankings, and neighborhood tiers, this analysis offers a data-driven view of how multiple neighborhood characteristics combine into distinct profiles. Specifically, it:

1. Identifies a smaller profile characterized by higher income and greater branch density, alongside a larger profile characterized by lower income and lower branch density.

2. Examines whether these profiles differ in account ownership, a related outcome excluded from the clustering inputs; and

3. Tests the stability of the resulting groups under PCA-based dimensionality reduction and alternative feature specifications.

Because Ward hierarchical clustering produced a stronger silhouette score than K-means, the Ward solution is compared with the NFHI classification of PUMAs as Advantaged, Vulnerable, or Variable as shown in Table B2.

Method	Cluster/Classification	Size
Ward	Economically advantaged (Cluster 1)	10
Ward	Economically disadvantaged (Cluster 0)	45
NFHI classification	Variable neighborhood	33
NFHI classification	Advantaged neighborhood	12
NFHI classification	Vulnerable neighborhood	10

Table B2. Ward clusters and NFHI classifications

The Ward method's economically advantaged cluster contains 10 PUMAs, closely approximating the 12 PUMAs classified by NFHI as Advantaged Neighborhoods. This similarity suggests that the clustering results may complement the NFHI tiers by providing an alternative, multivariate perspective on neighborhood conditions. However, the findings remain exploratory because the analysis includes only 55 PUMAs, cluster separation is modest, assignments are sensitive to feature selection, and the account-ownership comparison is descriptive rather than causal.

Discussion

What did you learn from doing Part A?

One surprising finding was that KNN outperformed both Ridge Regression and Random Forest, suggesting that PUMAs with similar socioeconomic characteristics may also have similar levels of physical banking access. Another interesting result was that removing homeownership slightly improved KNN's performance, showing that including more predictors does not always lead to better predictions.

One challenge was the small dataset of 55 NYC PUMAs, which made model performance sensitive to the training and validation splits. To address this, we used five-fold cross-validation to compare models more consistently. However, performance still varied across folds. Additionally, hyperparameter tuning and model evaluation used the same cross-validation setup, so the reported results may be somewhat optimistic.

The failure analysis also showed that socioeconomic characteristics alone cannot fully explain differences in banking access, particularly for PUMAs with unusually high branch density. With more time and resources, we could explore additional variables such as population density, commercial activity, and geographic accessibility. We could also use nested cross-validation to obtain a more reliable estimate of model performance.

What did you learn from doing Part B?

Some results surprised us. Ward method's cluster sizes line up with the NFH classification. The size of the smaller Ward cluster is close to the size of NFH classified advantaged neighborhoods. The Ward method's clusters align

with the financial health profiles. The smaller Ward cluster (10 PUMAs, mostly Manhattan plus a few Brooklyn and Queens neighborhoods) had a lower poverty rate (10.4% vs. 22.0%), higher median income ($61.6K vs. $25.9K), more branches per 10,000 residents (2.34 vs. 1.21), and higher held-out account ownership (0.934 vs. 0.862). Removing race-composition features changed the partition. ARI was 0.732. So, some PUMAs moved groups. The silhouette score also rose from 0.322 to 0.463 which shows that neighborhood profiles are partly sensitive to including race composition.

We encountered some challenges during the analysis. First, it was the small sample (55 PUMAs, 13 features). So, we limited k to 2-6, used n_init = 50 for K-means, and treated results as exploration. Second, features on very different scales and missing value. We standardized all features and median-imputed. To avoid circularity risk, Financial Services__Ind5Outcome (Bank / Credit Union Utilization) was kept out of the clustering inputs so it could serve as an independent check. To ensure cluster quality, we compared multiple metrics, visualizations, and sensitivity checks rather than relying on one score. To not overstate the validation, we described the bank/credit union utilization difference as descriptive, not causal.

If we have more time and resources, we could add more external data such as crime/complaint rate and number of employers/businesses and expand geographical areas and use more historical data if feasible. We could also try other clustering methods such as Gaussian models, DBSCAN, or HDBSCAN and compare results to additional similar research.

Ethical Considerations

Supervised Learning Ethical Considerations

The supervised model may reproduce structural inequalities because its socioeconomic predictors reflect historical segregation, discrimination, and unequal investment. Predictions of lower branch density must not be interpreted as evidence that a community needs or deserves less access, and the model should not automate branch placement or closure, lending, pricing, eligibility, or customer selection. Error patterns should be reviewed across boroughs, income levels, racial and ethnic composition, and NFH classifications.

Because the analysis uses only 55 PUMAs and aggregate neighborhood data, results are uncertain and cannot describe individual behavior, needs, or creditworthiness. Branch density measures physical access only; it excludes affordability, service quality, accessibility, digital banking, actual usage, and commercial activity. Results are exploratory—not causal or normative—and should support planning only alongside observed access, community input, transportation data, service-quality measures, and careful cross-validation.

Unsupervised Learning Ethical Considerations

Cluster results depend on feature selection, scaling, preprocessing, algorithm, and cluster count, so they should be presented as exploratory profiles rather than fixed neighborhood types. Race and ethnicity variables—and correlated socioeconomic proxies—may reproduce segregation and redlining. Solutions should therefore be compared with and without sensitive features, tested for stability, and kept out of lending, pricing, branch closure, or other high-stakes decisions.

Neutral labels should avoid stigmatizing communities or treating cluster numbers as rankings. PUMA-level clusters cannot characterize every resident or location, and median imputation or geocoding exclusions may bias representation if errors are concentrated in particular communities. Missingness and exclusions should be reported, alternative preprocessing choices tested, and clusters used only as a starting point for equity-oriented investigation with community input, transportation and digital access evidence, and human review.

Statement of Work

Team Member	Contributions
Sanika Babtiwale	Found project data sources, data preprocessing (PUMAs to bank locations, bank branch counts), notebook comments and cleanup, remaining report sections, team housekeeping (standup video uploads, maintaining the group Slack channel)
Zahra Makki	initial draft and data cleaning; found external data source; supervised learning section, both notebook code and written portion; overall final draft runthrough, attended standup video #1 and maintained consistent communication with team
Chachawal Thapanawat	Data cleanup (ratio-to-decimal conversion, flatten NFH data), Feature engineering (NFH Classification, ACS Population, Credit Union Branch Count), Part B - Unsupervised Learning, Standup meeting (Progress reports)

References

Federal Deposit Insurance Corporation. (n.d.). *BankFind Suite: Find institutions and locations*. Retrieved March 24, 2026, from https://banks.data.fdic.gov/bankfind-suite/locations

Federal Deposit Insurance Corporation. (2023). *2021 FDIC national survey of unbanked and underbanked households*. https://www.fdic.gov/analysis/household-survey/2021report.pdf

GeoPandas developers. (n.d.). *geopandas.points_from_xy*. GeoPandas documentation. Retrieved March 24, 2026, from https://geopandas.org/en/stable/docs/reference/api/geopandas.points_from_xy.html

Morgan, D. P., Pinkovskiy, M. L., & Yang, B. (2016, March 7). *Banking deserts, branch closings, and soft information*. Federal Reserve Bank of New York, *Liberty Street Economics*. https://libertystreeteconomics.newyorkfed.org/2016/03/banking-deserts-branch-closings-and-soft-information/

New York City Open Data. (n.d.). *Neighborhood financial health digital mapping and data tool* [Data set]. Retrieved March 24, 2026, from https://data.cityofnewyork.us/browse?q=Neighborhood%20Financial%20Health%20Digital%20Mapping%20and%20Data%20Tool

Nominatim contributors. (n.d.). *Search API*. Nominatim documentation. Retrieved March 24, 2026, from https://nominatim.org/release-docs/develop/api/Search/

OpenStreetMap contributors. (n.d.). *OpenStreetMap*. Retrieved March 24, 2026, from https://www.openstreetmap.org/copyright

OpenStreetMap Foundation. (n.d.). *Nominatim usage policy*. Retrieved March 24, 2026, from https://operations.osmfoundation.org/policies/nominatim/

Ratcliffe, C., Buron, L., Stanczyk, A., Martín, C., & Kotapati, B. (2015). *Where are the unbanked and underbanked in New York City?* Urban Institute. https://www.urban.org/research/publication/where-are-unbanked-and-underbanked-new-york-city

U.S. Census Bureau. (2023). *2023 Gazetteer files: ZIP Code Tabulation Areas* [Data set]. https://www2.census.gov/geo/docs/maps-data/data/gazetteer/2023_Gazetteer/2023_Gaz_zcta_national.zip

U.S. Census Bureau. (n.d.-a). *Census geocoder*. Retrieved March 24, 2026, from https://geocoding.geo.census.gov/geocoder/

U.S. Census Bureau. (n.d.-b). *Census geocoder: Find geographies using a batch of addresses*. Retrieved March 24, 2026, from https://geocoding.geo.census.gov/geocoder/geographies/addressbatch

U.S. Census Bureau. (n.d.-c). *Census geocoder: Find geographies using coordinates*. Retrieved March 24, 2026, from https://geocoding.geo.census.gov/geocoder/geographies/coordinates

U.S. Census Bureau. (n.d.-d). *Census geocoding services API*. Retrieved March 24, 2026, from https://geocoding.geo.census.gov/geocoder/Geocoding_Services_API.html

U.S. Census Bureau. (n.d.-e). *Gazetteer files*. Retrieved March 24, 2026, from https://www.census.gov/geographies/reference-files/time-series/geo/gazetteer-files.html

U.S. Census Bureau. (n.d.-f). *TIGER/Line shapefiles*. Retrieved March 24, 2026, from https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html